MASooda — Team Report: Proposal vs. Delivered Work

Nov 10, 2026 · @Abdur Rehman

Summary

MASooda delivered all six outcomes promised in the proposal and went well past its scope: a live road-corridor simulator over the IIT Hyderabad campus, a deployed browser demo, a battery-management (BMS) layer and an IEEE-style paper draft.

- **Proposal (Sep 2026):** a 2D/2.5D sim of 50–80 drones on 2–3 corridors, with boids, ADMM deconfliction, a priority-aware overseer, W-MSR resilience against one attack, one live STL spec and a three-way architecture comparison.
- **Delivered (Nov 2026):** weekly modules for every role (Weeks 1–6), then a second simulator (`simulation/route_engine.py`) on traced campus and Hyderabad roads with 7 altitude layers and fleets up to 500 drones, which drives the final demo `python -m web_sandbox.server` (live at masooda.onrender.com) and the paper.
- **Beyond scope:** drone health and BMS (1-RC battery, EKF state of charge, chance-constrained energy), pad arbitration, energy-weighted ADMM, layer agreement, an operator console, 10 scripted scenarios and a 460-run paper evaluation.
- **Who:** R Lalith Vishnu (P2) built the route-engine simulator, the final demo, the BMS work and the paper, and owns the ADMM module. Git history shows 57 of 106 commits under his two identities (`rlv-iith` , `R Lalith Vishnu`); a few of his P2 and Week 6 deliverables were committed by teammates (see Attribution notes).

Source: the repository's git history on all branches (current branch `bms/publishable` , head `22c5400`), `git blame` on tracked code, the proposal and `workflow.pdf` .

Team and roles as proposed

The proposal split the work into six roles; P1 and P6 were shared by Abdur and Dhadheechi.

Role	Proposed owner	Proposed deliverable	Code area
P1 Simulation engine	Abdur + Dhadheechi	Agent dynamics, comm graph, renderer, integration	`simulation/`
P2 Strategy and conflict resolution	Lalith	Distributed ADMM deconfliction	`conflict_resolution/`
P3 Local behaviour and formation	Bhuvan	Boids rules, corridor merging, platooning	`local_behavior/`
P4 Global overseer and priority	Tanmay	Corridor allocation, priority arbitration, graceful degradation	`simulation/overseer*.py`
P5 Adversarial resilience	Sampad	Threat model, attack injection, W-MSR	`resilience/`
P6 Formal verification and demo	Abdur + Dhadheechi	STL specs, robustness monitor, dashboard	`verification/`, `visualization/`

Proposal outcomes vs. delivered

All six promised outcomes were delivered. Two items the proposal listed as stretch goals (500+ agents and a second STL spec for reachability) were also reached.

#	Proposed outcome	What was delivered	Where
1	Live sim of N heterogeneous drones in corridors, with an overseer	Weeks 1–5: grid engine with 2.5D altitude layers. Final: road-corridor engine on 28 traced IITH roads (105 junctions, 13 vertiports) and Hyderabad OSM roads, 7 layers, fleets up to 500, streamed to the browser	`simulation/engine.py`, `simulation/route_engine.py`, `web_sandbox/`
2	Priority arbitration across drone classes	Emergency > eVTOL > delivery in overseer allocation and ADMM weights (10/3/1); reserved emergency layer; emergency-delay and on-time KPIs; medical-emergency preset	`simulation/overseer.py`, `simulation/route_priority.py`
3	Resilience to adversarial agents	Position-spoofing attack, trusted-range check, W-MSR (f = 1); paper run: 0 spoofed reports accepted vs. 193 per run undefended	`resilience/`

#	Proposed outcome	What was delivered	Where
4	Hierarchical vs. centralized vs. decentralized	Four architectures from identical seeds (hierarchical, centralized, decentralized, undefended); Week 5 E4 and paper Table 7	`simulation/centralized.py`, `simulation/route_centralized.py`
5	Formal safety guarantee (STL)	G(d ≥ d_min) with live robustness ρ(t), layer-aware; optional F(reach destination) spec; near-miss and crash detection	`verification/`
6	Graceful degradation without the overseer	Overseer link failure (Week 4), E6 degradation experiment, DTC outage preset; drones keep last directive and queue commands	`simulation/overseer_link.py`, `route_engine.py`

Added beyond the proposal (all in the final demo and paper):

- Drone health and BMS: 1-RC Thevenin pack with per-cell spread, thermal model, EKF on (SoC, V_RC, 1/Q), chance-constrained usable energy, voltage-sag check, scheduled faults.
- Health-aware traffic management: endurance-ordered pad arbitration, hold-short, energy-margin ADMM weights.
- Layer agreement: V2V layer negotiation plus DTC congestion pricing across all layers.
- DTC operator console (reroute, hold, return to base, ground stop, TFR zones, quarantine, battery swap), weather and wind, 10 scripted scenario presets, run export and replay.
- Paper: 460 paired-seed runs with a second-platform reproducibility check.

Not done, as the proposal allowed: true 3D rendering, several simultaneous attack types, full epistemic-logic belief modelling and learned (RL) policies.

Who did what

Every member delivered their own module; Lalith also built the second simulator, the final demo, the BMS layer and the paper. Commit counts are on all branches; files are named by the module each person wrote first, as `git blame` shows them now.

Member	Role	Commits (all branches)	What they built (evidence)
R Lalith Vishnu (`rlv-iith`, `R Lalith Vishnu`)	P2 + final demo, simulator, BMS, paper	57	**P2 ADMM:** solver, constraints, convergence, adapter, conflict detection, N-agent ADMM, profiler and Week 2–5 experiments (`conflict_resolution/`). **Road-corridor simulator:** `road_network.py`, `route_engine.py` (2,169 lines), `spatial_grid.py`, `route_formation.py`, `route_priority.py`, `route_centralized.py`, `admm_probe.py`, `layer_agreement.py`. **Maps:** IITH campus (traced roads, geofence, 13 vertiports) and Hyderabad OSM. **BMS:** `drone_health.py` (1-RC battery, EKF, chance-constrained energy, pad arbitration). **Final demo:** `web_sandbox/` (server, streaming, export, front end), `dtc_console.py`, `scenario_presets.py`, `scenario_manifest.py`, Render deployment. **Earlier demos:** `final_demo.py`, 2.5D ADMM demo, SQLite run storage. **Docs:** project audit, integration plan, research gap, live demo, drone health, layer agreement, scenario presets. **Week 6:**

Member	Role	Commits (all branches)	What they built (evidence)
			experiments, plots, dashboard, `run_all`, reproducibility (see Attribution notes). **Paper:** manuscript and 460-run evaluation.
Sampad (`Sampadram`)	P5 Resilience; weekly integrator	28	Threat model, attack injection, standard consensus baseline, W-MSR and report defence (`resilience/`), decentralized-undefended baseline, E3/E4/E6 Week 5 experiments, priority metrics, 2.5D altitude layers in the grid engine, STL monitor and toy trajectories (Week 1 integration), weekly merges of every role's branch (Weeks 1–5).
Mohammed Abdur Rehman (`MohammedAbdurRehman11015`)	P1 + P6	11	Grid simulation engine and loop (`engine.py`), comm graph, environment, scenario, logger, telemetry, centralized architecture (Week 5), live STL monitor, metrics and near-collision checks (`verification/`), renderer extensions and dashboard/plots (`visualization/`), Week 4 MVP run. The Week 6 commit is under his name (see Attribution notes).

Member	Role	Commits (all branches)	What they built (evidence)
Dhadheechi	P1 + P6	1	P1 Week 1: agent dynamics (`dynamics.py`), shared schema, mock data, uv setup and the top-down renderer.
Bhuvan	P3 Formation	4	Boids alignment, cohesion and separation (`local_behavior/`), corridor selection and merge-gap control, local-behaviour cache reset and ADMM merge check, Week 5 formation metrics and sweep.
Tanmay Banjari	P4 Overseer	5	Overseer contract and skeleton, priority-aware corridor allocation (Week 3), overseer link failure and graceful degradation (`overseer_link.py`, Week 4).

The final demo and the paper run on Lalith′s road-corridor engine, which imports the team′s modules: Dhadheechi′s dynamics, Bhuvan′s boids, Tanmay′s overseer and link model, Sampad′s attack and W-MSR defence, Abdur′s comm graph and STL metrics, and the P2 ADMM solver.

Attribution notes

Some of Lalith′s work reached the repository through teammates′ commits. Lalith states he did this work; the table names each commit so the team can confirm it.

Commit	Date	Git author	Lalith's work in it
`71108da` week 6 done	7 Nov	Abdur	Week 6: `experiments/week6*.py`, E1–E6 run and plot scripts, `run_all`, dashboard, telemetry, `docs/REPRODUCIBILITY.md`, `DEMO_SCRIPT.md`, `DRY_RUNS.md`, `DOD_AUDIT.md` and Week 6 tests (about 7,600 lines of code now in the tree)
`e24249b` week 4 p5	1 Nov	Sampad	P2 Week 4 ADMM profiling (`scripts/p2_week4_profile.py`, `docs/p2_week4_profile.md`)
`8fc3e47` Week 3 work p5	1 Nov	Sampad	P2 Week 3: conflict detection, N-agent ADMM, priority tests (`conflict_resolution/detection.py`, `scripts/p2_week3_nagent.py`, `tests/test_p2_week3_*.py`)
`17d41a0` Week 2 work	1 Nov	Sampad	P2 Week 2: ADMM adapter and iteration sweep (`conflict_resolution/adapter.py`, `scripts/p2_week2_iteration_sweep.py`, `tests/test_p2_week2.py`)
`c75bb8f` Week 1 integration	1 Nov	Sampad	P2 Week 1: two-agent ADMM solver (`conflict_resolution/solver.py`, `constraints.py`, `convergence.py`, `tests/test_admm_solver.py`); the commit message names "P2 ADMM"

The same Sampad commits also carry Sampad's own P5 work, so only the P2 files above move. Counting them, P2 code under `conflict_resolution/` and the P2 scripts, tests and docs is about 2,300 lines, of which `git blame` shows about 1,660 under Sampad's name.

Open question: are there other commits (for example any P1 or merge work) that should be listed here?

Final demo: `web_sandbox.server`

The final demo is `python -m web_sandbox.server` (port 8765), deployed at masooda.onrender.com. Lalith wrote all of `web_sandbox/` (about 3,200 lines across server, streaming, export and front end) and the route engine behind it.

How a run flows:

1. The browser picks a map (IITH campus or Hyderabad), a fleet size up to 500 or a scenario preset, and an architecture.
2. `server.py` validates the request and runs `route_engine` per tick: dynamics, comm graph, boids, ADMM on conflict groups, DTC layer allocation and layer agreement, attack and W-MSR defence, BMS and onboard health responses.
3. `streaming.py` sends compact frames with an adaptive stride; the browser draws them on Canvas 2D.
4. Metrics (STL robustness, near misses, emergency delay, ADMM timer) update live; the run downloads as JSON for replay.

What the room sees, against the proposal's demo vision:

Proposal's demo beat	In the final demo
Normal operation, swarm in corridors	Drones on road corridors over 7 layers, sector and layer occupancy, 2.5D side view
Malicious drone plus overseer link drop	GPS-spoofing attack with reported vs. estimated positions; DTC outage preset
Emergency drone gets right of way	Medical-emergency preset, reserved emergency layer, emergency-delay KPI
Comparison chart	Side-by-side variants and the four-architecture comparison from identical seeds

Extra panels: battery matrix of the whole fleet, per-drone BMS and cell voltages, V2V network panel, weather and wind, and the DTC operator console. The server was also hardened for public use: one visitor's launch no longer stops another's run, and downloads are built in the browser.

Paper draft

Lalith wrote the draft *BMS-in-the-Loop Traffic Management for Urban Drone Fleets: Chance-Constrained Energy Reserves, Pad Arbitration, and Energy-Weighted ADMM Deconfliction* (11 pages, target IEEE Access), with every result produced from branch `bms/publishable` (head `22c5400`). Authors: R Lalith Vishnu, Sampadram, Mohammed Abdur Rehman, Bhuvan, Tanmay Banjari, Dhadheechi.

- **Covers:** the whole hierarchical system as the base, with the BMS layer as the contribution: 1-RC pack model, EKF on (SoC, V_RC, 1/Q), chance-constrained energy with z = 1.645, pad arbitration and energy-margin ADMM weights (with a proof of their bounds).
- **Evaluation:** 460 paired-seed runs, 100 drones on the campus map; 53 runs repeated on a second platform with identical discrete outcomes.
- **Main results:** SoC error below 1.0% in nominal flight and never an over-estimate of usable energy; in a low-charge fleet the chance constraint kept 6.4 more drones on the ground and avoided 3.5 in-flight diversions per run, with deliveries unchanged; the hierarchical design had 0 near misses vs. 9.8 per run for the centralized baseline.
- **Honest limits stated:** health-aware arbitration and weighting gave no measurable gain in the tested scenarios, one seed produced a forced landing from hold-short placement, and the threshold thermal response was too slow at 44 °C.
- **Package:** `paper/` holds the LaTeX source, figures, 14 experiment and analysis scripts, raw results (`runs_final.jsonl`) and tests. It is not tracked in git yet.

The paper excludes the Week 6 results and does not report the earlier three-corridor engine's experiments, which it says were superseded.

Timeline and git statistics

Most of the code landed between 1 and 10 November; Lalith's commits from 5 November onward produced the final simulator, demo, BMS and paper.

Date (2026)	Milestone	By
10 Nov	1-RC battery and EKF, chance-constrained energy, pad arbitration, energy-margin ADMM weights, paper experiments	Lalith

Date (2026)	Milestone	By
9 Nov	Layer agreement (V2V + DTC pricing), 7 layers, demo redesign	Lalith
7 Nov	Live demo phases 0–5: scaling to 500, integration, custom drones, drone health, DTC console, presets; deployed to Render	Lalith
7 Nov	Week 6 experiments, plots, dashboard, `run_all` (committed by Abdur)	Lalith (per Lalith)
6 Nov	Road-corridor engine, IITH campus and Hyderabad maps, project audit, ADMM fixes	Lalith
5 Nov	Week 5: centralized baseline, E1–E6 experiments, `final_demo.py`	All roles
4 Nov	2.5D altitude layers in the grid engine	Sampad
1 Nov	Weeks 1–4 committed and integrated: boids, ADMM, overseer, W-MSR, STL, MVP	All roles
30 Sep	P1 Week 1 simulation engine	Dhadheechi
15–20 Sep	Proposal, README, P4 overseer contract	Lalith, Tanmay

Code now in the tree (Python, JS, HTML, CSS outside `docs/` and `outputs/`; 193 files), by `git blame`:

Member	Lines as committed	Lines after attribution notes
Lalith (both identities)	12,400	about 21,600
Abdur	12,023	about 4,460
Sampad	4,986	about 3,330
Bhuvan	2,374	2,374
Dhadheechi	1,102	1,102
Tanmay	1,050	1,050

The second column moves the Week 6 commit (about 7,560 lines) and the P2 files under Sampad's name (about 1,660 lines) to Lalith. Tests: 421 test functions in 80 files.